Fiscal Year:	FY 2007	**Task Last Updated:**	FY 01/28/2008
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Project Title:	The Genetic Basis of the Loss of Musculo-Skeletal Tissue during Weightlessness: Towards the Identification of Individuals that are at Greatest Risk		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Physiology		
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Zip Code:	11794-5281	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	01-OBPR-07
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Contact Monitor:		**Contact Phone:**	
Contact Email:			
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Flight Assignment:	NOTE: Received NCE to 7/26/07 per PI (1/07) NOTE: Received NCE to 1/26/2007 per J. Dardano (5/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Donahue, Leah Rae (The Jackson Laboratory) Rubin, Clinton (SUNY Stony Brook)		
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Task Description:

Task Abstract/Description (From Original Grant Proposal):
The National Research Council's Space Studies Board has stated that a principal physiologic hurdle to man's extended presence in space is the osteopenia and sarcopenia which parallels reduced gravity. The extent of the loss is extremely high, approaching a decrease in bone mineral density (BMD) in the lower appendicular skeleton at a rate of 1.6% per month and reducing maximal voluntary contractions of some muscle groups at a rate of 5% per month. Interestingly, the amount of bone (and muscle) loss between individual astronauts is highly variable with some astronauts losing large amounts of tissue while others are largely unaffected. This large individual variability, which has also been observed during bedrest and immobilization studies on Earth, may be accounted for, at least to a large extent, by genetic variations which may give rise to a differential mechanosensitivity of the musculoskeleton. We have collected preliminary data demonstrating that genetically distinct inbred strains of mice also demonstrate a distinct sensitivity to conditions of simulated weightlessness; while as much as 60% of trabecular bone is lost in the hindlimbs of BALB/cByJ mice within 3 weeks of disuse, the same conditions leave trabecular bone quantity and quality nearly unchanged in C3H/HeJ mice. In this proposal, we aim to elucidate this genetic basis of the sensitivity of the musculo-skeleton to the loss of appropriate mechanical signals by identifying the quantitative trait loci (QTL) responsible for the difference exhibited between these two strains of inbred mice. The identification of QTL (and ultimately the responsible genes) may be used as both a critical diagnostic sensor for the identification of astronauts that are in greatest need of pharmacologic and/or biomechanical countermeasures in space and as a discovery tool of novel drug targets for the preventing the loss of musculo-skeletal tissue.

Task Objectives:

* to reveal the chromosomal locations (genes) that are associated with a high and low sensitivity of bone and muscle to hindlimb unloading and reambulation by double-crossing a mechanically responsive (BALB/cByJ) and a mechanically unresponsive (C3H/HeJ) mouse

* to study whether the tight association between muscle and bone will cause QTLs for muscle to overlap with QTLs for bone.

* to develop a genetic diagnostic sensor to identify individuals who at highest risk of losing bone and muscle due to space flight, bedrest, or aging.

* to link the three-dimensional changes in bone architecture induced by unloading/reambulation to micro finite element models to identify changes in the mechanical properties of trabecular and cortical bone.

* to probe the influence of genetics on the recovery of bone loss upon a period of unloading.

* to serve as a discovery tool for novel drug targets that will prevent the loss of musculo-skeletal tissue.

Task Significance:

The very large variability in bone loss between astronauts (i.e., some lose a lot, some lose almost nothing), in the absence of large variations in environmental factors (e.g., diet or exercise), suggest a strong genetic influence on bone's susceptibility to the loss of functional weightbearing. The principal aim of this proposal to elucidate the genetic basis of the sensitivity of the musculo-skeleton to the loss of appropriate mechanical signals may lead to the design of a diagnostic sensor for the identification of astronauts that are in greatest need of pharmacologic and/or biomechanical countermeasures in space, as well as to the discovery of novel drug targets that will allow for safe and efficacious countermeasures.

This proposal addresses two questions that are central to the Critical Path:

Relevance to Critical Path:

Question 1g: What are the important predictors for estimating site-specific bone loss and fracture risk during hypogravity exposure, including, but not limited to ethnicity, gender, genetics, age, baseline bone density and geometry, nutritional status, fitness level and prior microgravity exposure?

The focus of this grant is to investigate how genetics modulates bone quantity and micro-architecture and, more specifically, to identify the specific chromosomal regions that predispose individuals to the loss of muskulo-skeletal tissue in space. Knowledge of these chromosomal regions (QTL) may be used as a diagnostic tool before space travel (and the loss of tissue) and for the design of individually tailored treatment regimes. Further, as described below, we are also investigating how gender, baseline density and geometry affect site-specific bone loss.

Question 1k: Are hypogravity-induced changes in bone density, geometry, and architecture reversible upon encountering partial gravity exposure, or on return to full gravity (1-G)?

Mice that are subjected to hindlimb suspension in this study are allowed, upon completion of 3wk of unloading, to resume normal cage activities and any restoration in bone morphology and architecture is assessed by in vivo micro-computed tomography.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Skeletal pathologies involving bone loss, such as osteoporosis, are having a catastrophic impact upon the quality of life of the affected population and upon the cost of North American health care. In the United States alone, it is estimated that more than 44 million women and men (that is 55% of all people older than fifty years) are inflicted with osteoporosis or will be at high risk of developing it, currently costing the health care system more than $20 billion annually. Data derived from these studies will not only be relevant to spaceflight but be equally important to the prevention and treatment of bone loss associated with bedrest, musculo-skeletal diseases, and decreased levels of physical activity in our aging population.

Task Progress:	A genetically heterogeneous mouse population was created by double crossing two inbred strains of mice with very distinct skeletal responses to the loss of functional weightbearing. More than 400 of these mice were subjected to hindlimb unloading and reambulation. Longitudinal changes in the morphology and mechanical properties of these mice were determined by in vivo microCT and finite element modeling. Similar to data from astronauts and bedrest patients, changes in bone morphology during disuse and reambulation were associated with large variations in this genetically heterogeneous mouse population. The loss of bone quantity and quality during disuse was particularly dramatic in trabecular bone. Even though most mice were able to recover at least some of the tissue mass that was lost during disuse, this recovery occurred by re-thickening of individual trabeculae while the connectedness of the structure further deteriorated. The amount of tissue recovery was neither associated with any baseline bone morphology parameter nor with the rates of tissue loss induced by disuse. These data indicate the strong influence of genetic variations on the changes in bone morphology and strength induced by rapid changes in its mechanical milieu, emphasize that is critical to preserve bone's micro-architecture before individual trabecular connections are lost, and suggest that the genes modulating changes in morphology during disuse are, at least in part, distinct from those regulating tissue recovery. The weak correlations between bone and muscle loss and recovery further suggest that there is little overlap between the mechanosensory elements of muscle and bone. The phenotyping and genotyping of all mice is complete and the current identification (expected completion: March 2008) of the specific sets of QTL underlying the tissue-level observations may allow the identification of those individuals that are most susceptible to bone loss and the least capable of recovering.
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